

Patient Profile

Yash M. Patel, 21 years, gender unspecified

Summary of Results

The predictive model assessed the overall profile as **within normal limits**.

However, clinical context suggests possible non-alcoholic fatty liver disease (NAFLD) with reported elevations of AST and ALT, though specific numerical values were not provided.

Detailed Test Explanations

- **AST (Aspartate Aminotransferase):** Enzyme present in liver and cardiac muscle. Elevated levels indicate hepatocellular injury or inflammation.
- **ALT (Alanine Aminotransferase):** Liver-specific enzyme; higher concentrations are more specific for hepatic injury than AST.

Normal interpretation of these enzymes depends on laboratory reference ranges, typically < 40 U/L for each. In the absence of documented values, the elevations are presumed to be clinically significant but unquantified.

Overall Interpretation

The machine-learning assessment of the cumulative profile—including patient demographics and the mention of elevated hepatic enzymes—suggests no immediate red flag that would warrant an urgent therapeutic intervention. Clinically, nonetheless, the elevated AST and ALT, together with the suspicion of NAFLD, indicate subclinical hepatic involvement that should be monitored.

Recommendations

- **Lifestyle Modification:** Adopt a Mediterranean-style diet low in saturated fats and simple sugars; aim for regular aerobic exercise (≥ 150 min/wk).
- **Weight Management:** Target a body mass index (BMI) of 18.5–24.9 kg/m² to reduce hepatic fat deposition.
- **Alcohol Avoidance:** Strict abstinence from alcohol, as it can exacerbate hepatic steatosis.
- **Routine Monitoring:** Re-check liver enzymes (AST, ALT) and imaging (ultrasound or FibroScan) every 6–12 months, or sooner if symptoms develop.
- **Medical Follow-up:** Consider referral to a hepatologist or gastroenterologist for further evaluation and possible non-invasive fibrosis assessment.

Disclaimer

This report is informational and not a substitute for personalized medical advice. Please consult a qualified healthcare provider for diagnosis and treatment.